

DEPARTMENT OF COMPUTER SCIENCE

LAB1

Submitted by:

Mehmoona Abid

Submitted to:

Sir Waqas

Roll no:

BCS-2023-097

Semester:

BCS 7A

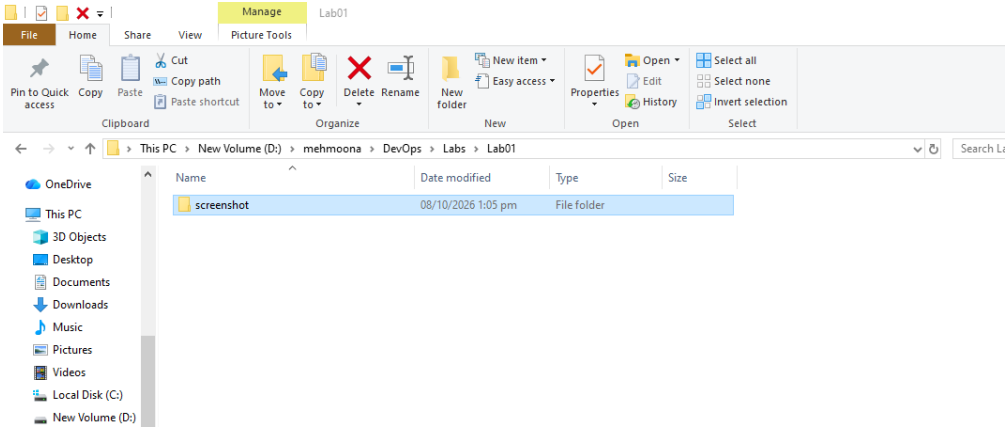
GitHub Username:

mehmoonaabid

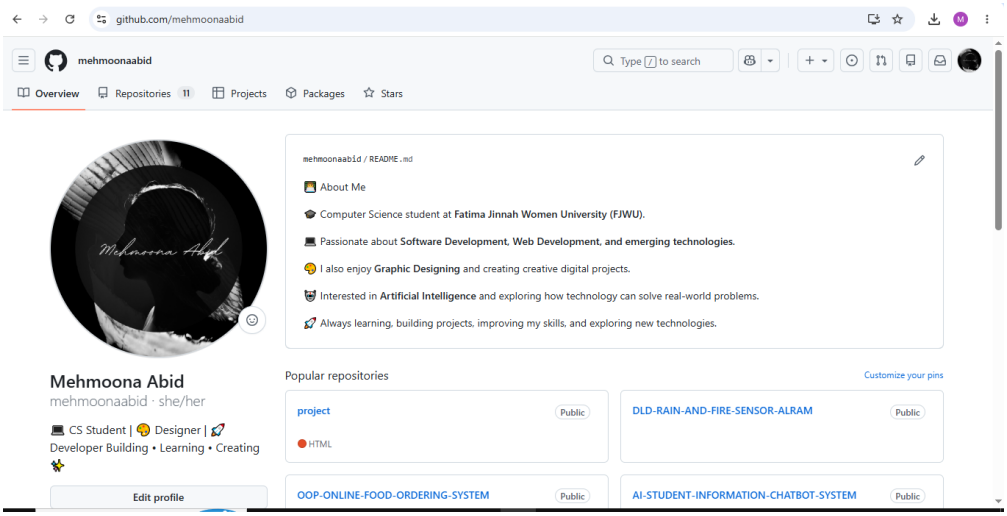
Lab 01:

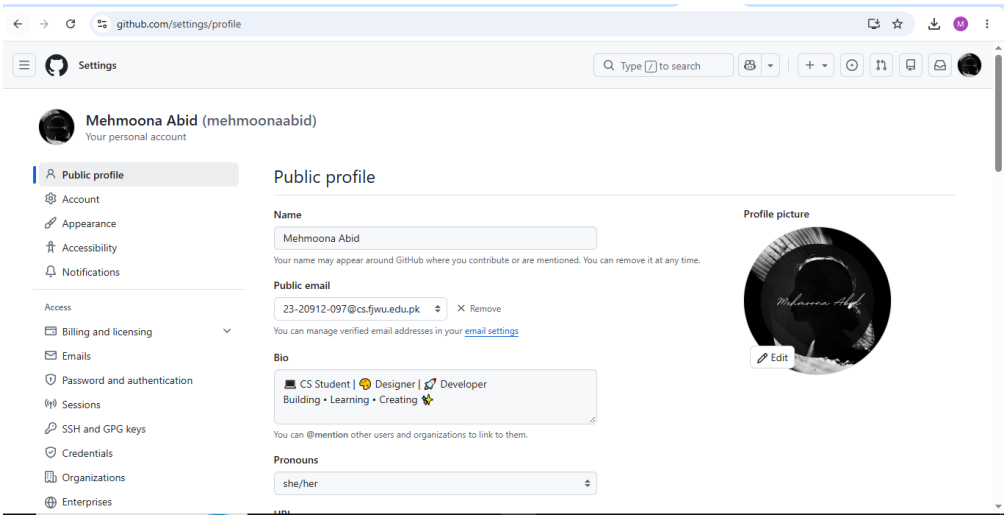
GitHub Identity, Student Portal Registration, and Ubuntu Server Installation

Getting started folder structure

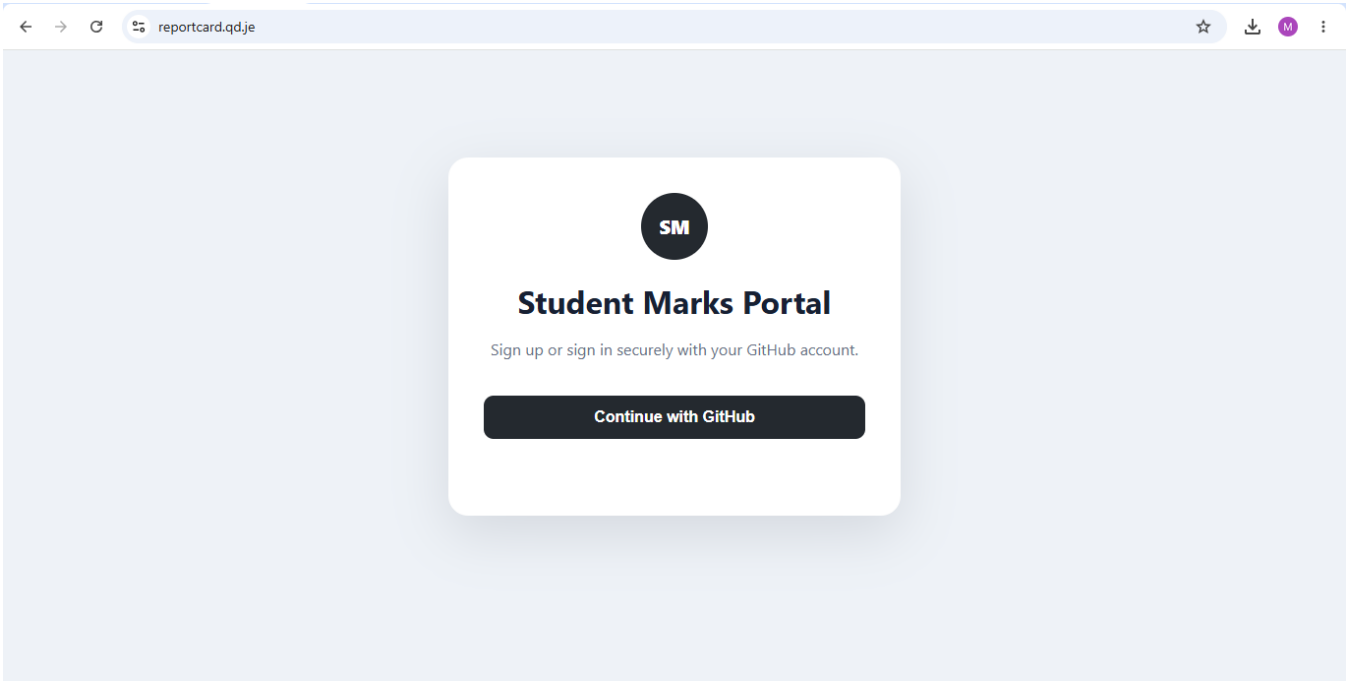


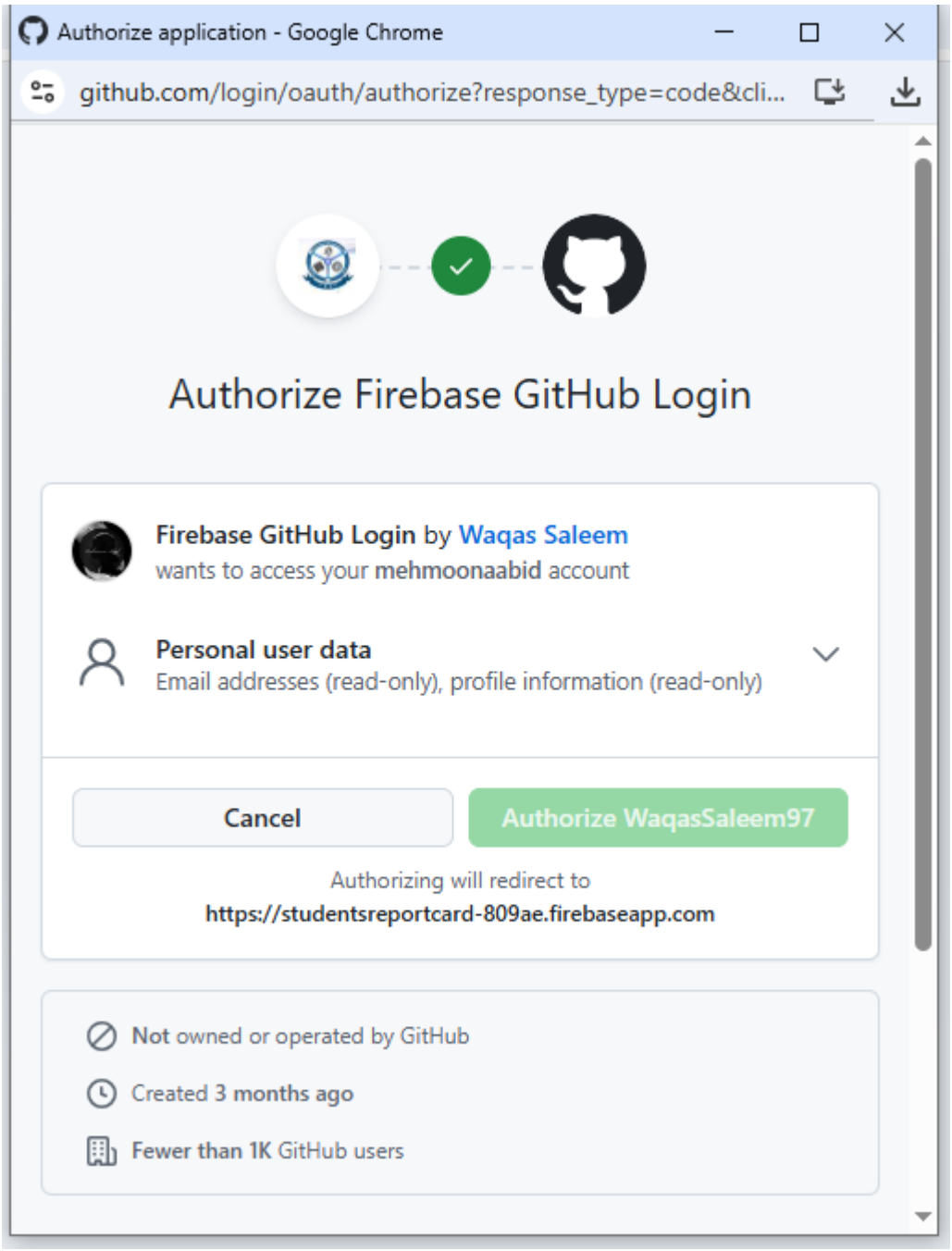
Task 1: Create and Configure Your GitHub Account

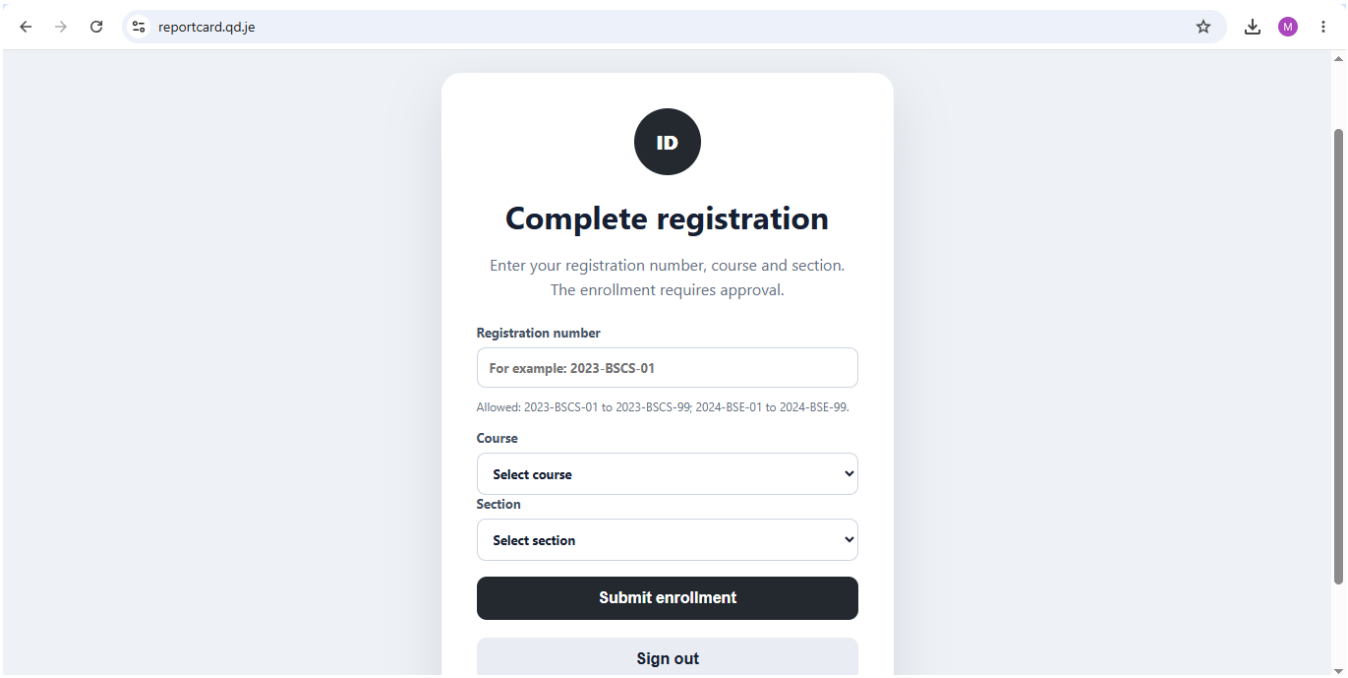


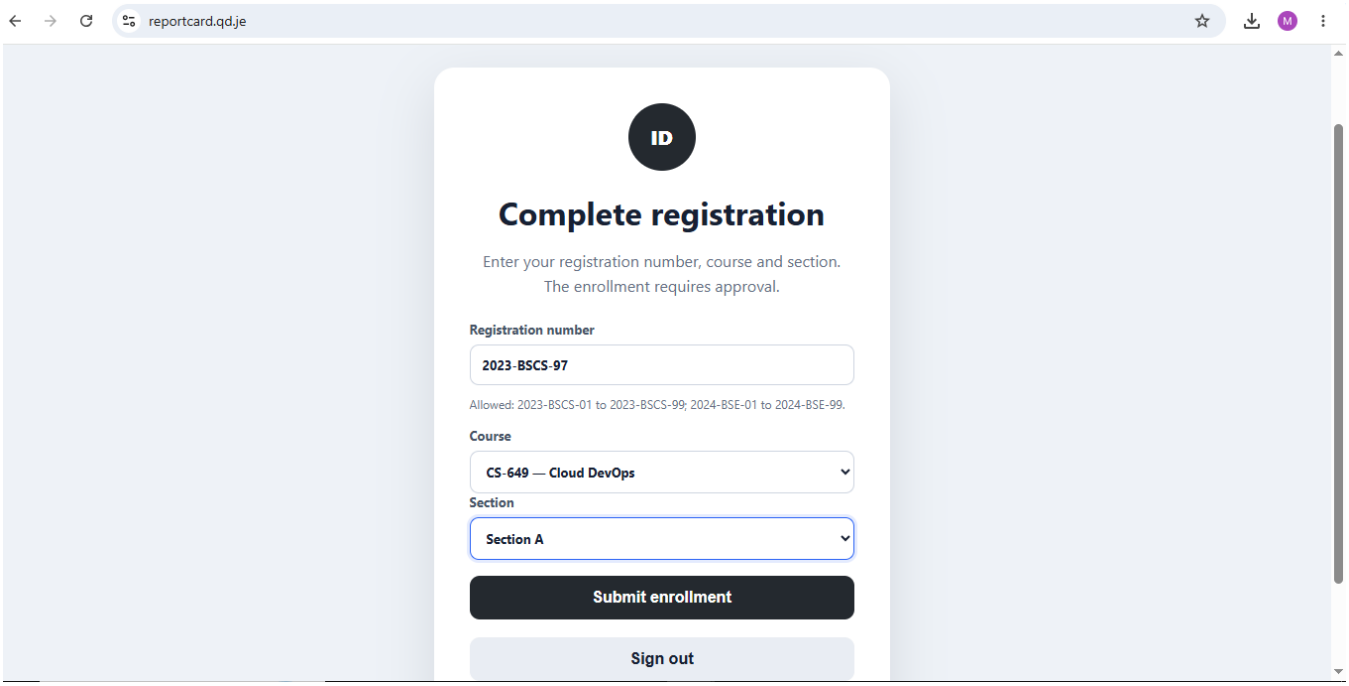


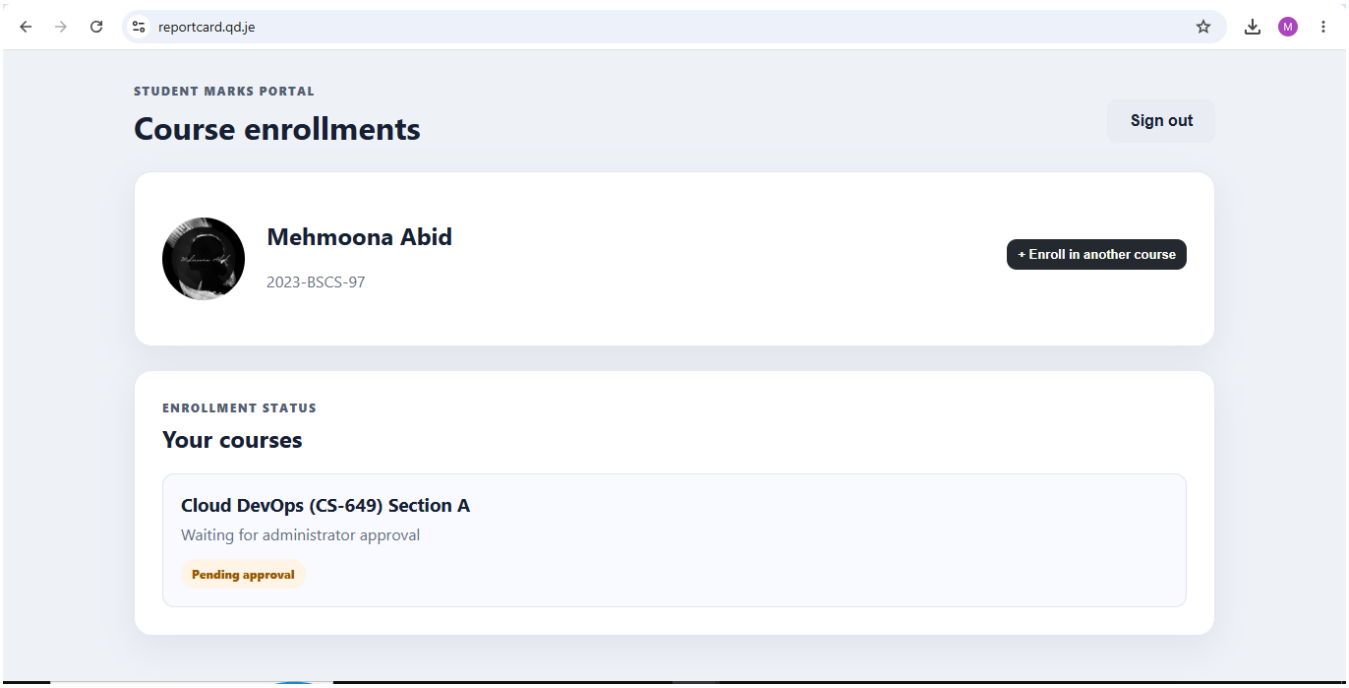
Task 2: Register in the Student Marks Portal



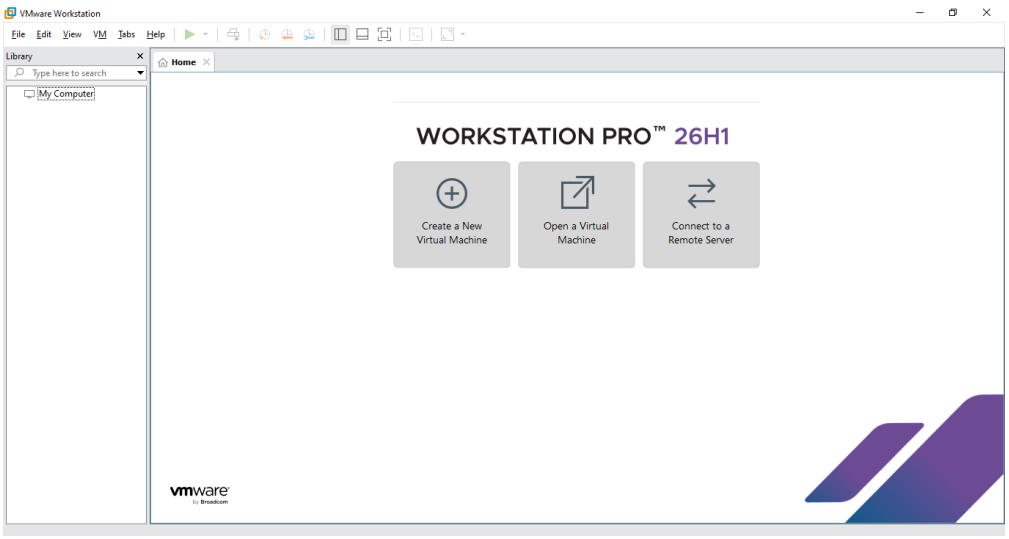

Authorize application - Google Chrome
github.com/login/oauth/authorize?response_type=code&cli...
Authorize Firebase GitHub Login
Firebase GitHub Login by Waqas Saleem
wants to access your mehmoonaabid account
Personal user data
Email addresses (read-only), profile information (read-only)
Cancel
Authorize WaqasSaleem97
Authorizing will redirect to
https://studentsreportcard-809ae.firebaseapp.com
Not owned or operated by GitHub
Created 3 months ago
Fewer than 1K GitHub users

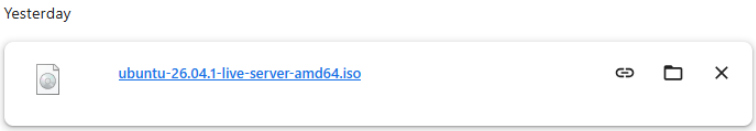

reportcard.qd.je
ID
Complete registration
Enter your registration number, course and section.
The enrollment requires approval.
Registration number
For example: 2023-BSCS-01
Allowed: 2023-BSCS-01 to 2023-BSCS-99; 2024-BSE-01 to 2024-BSE-99.
Course
Select course
Section
Select section
Submit enrollment
Sign out

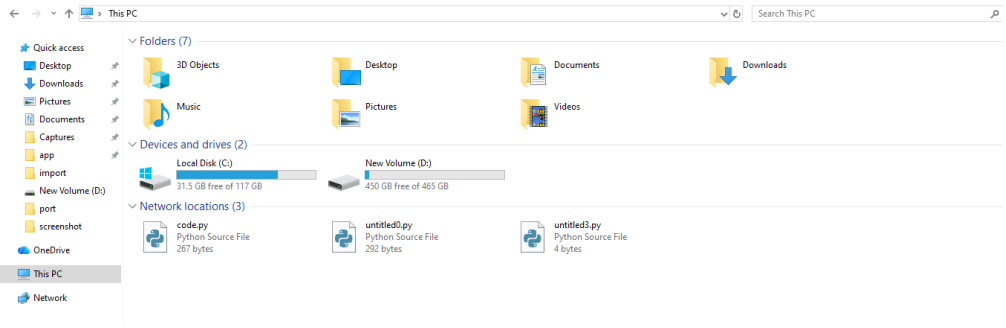

reportcard.qd.je
ID
Complete registration
Enter your registration number, course and section.
The enrollment requires approval.
Registration number
2023-BSCS-97
Allowed: 2023-BSCS-01 to 2023-BSCS-99; 2024-BSE-01 to 2024-BSE-99.
Course
CS-649 — Cloud DevOps
Section
Section A
Submit enrollment
Sign out



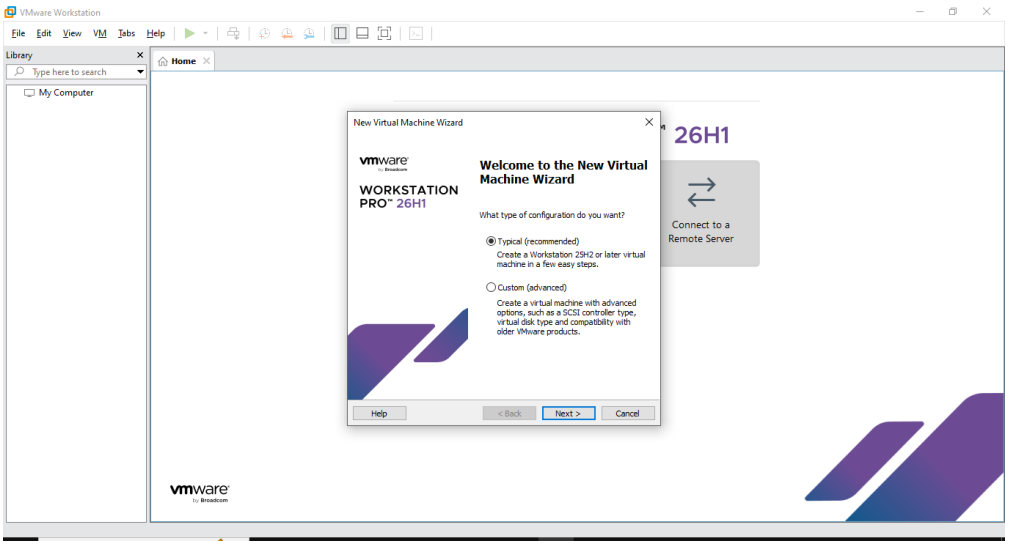
Task 3: Download VMware Workstation Pro and Ubuntu Server

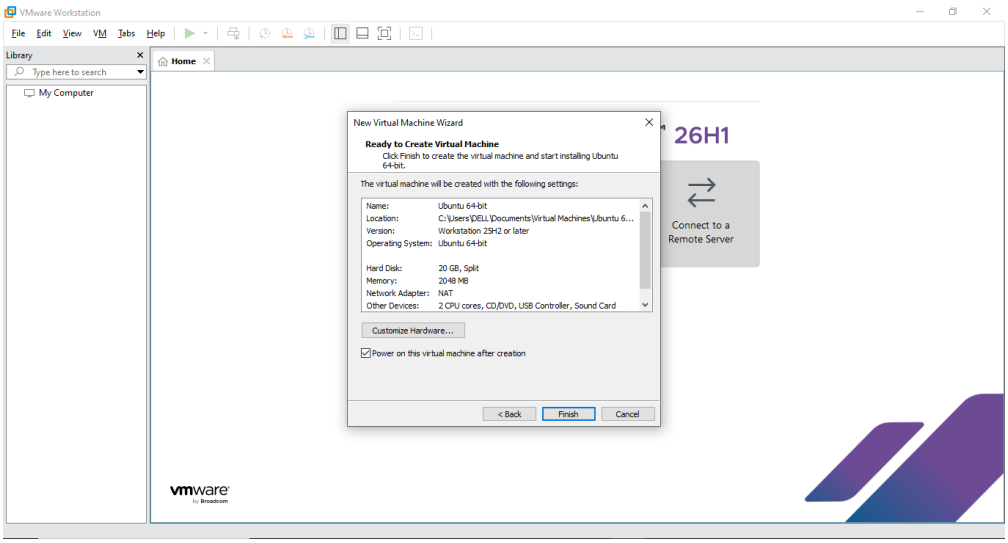


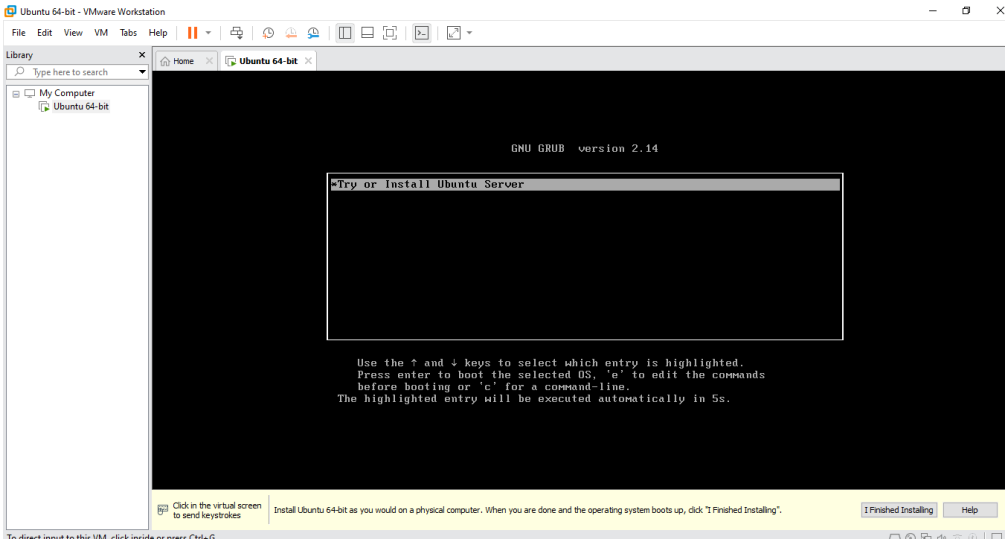




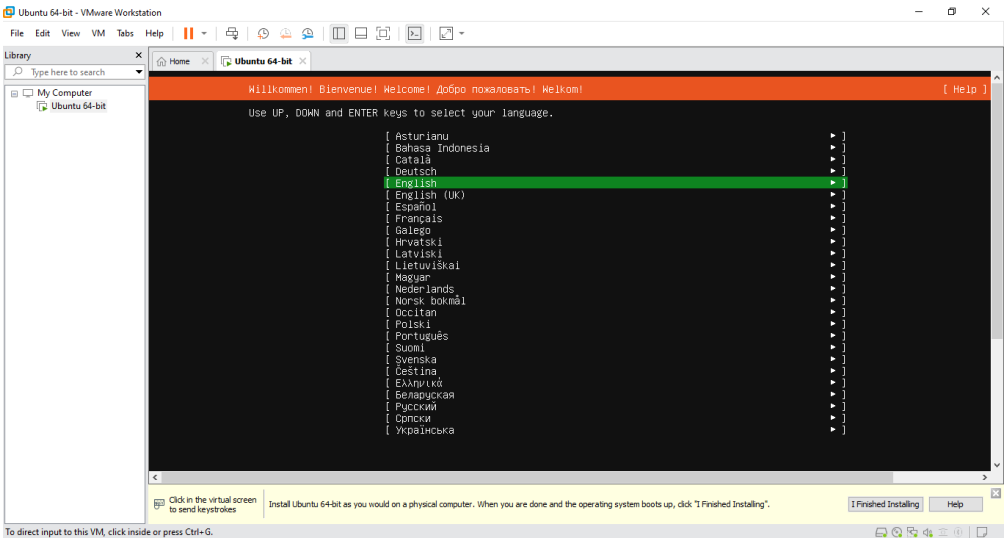
Task 4: Create the Ubuntu Server Virtual Machine



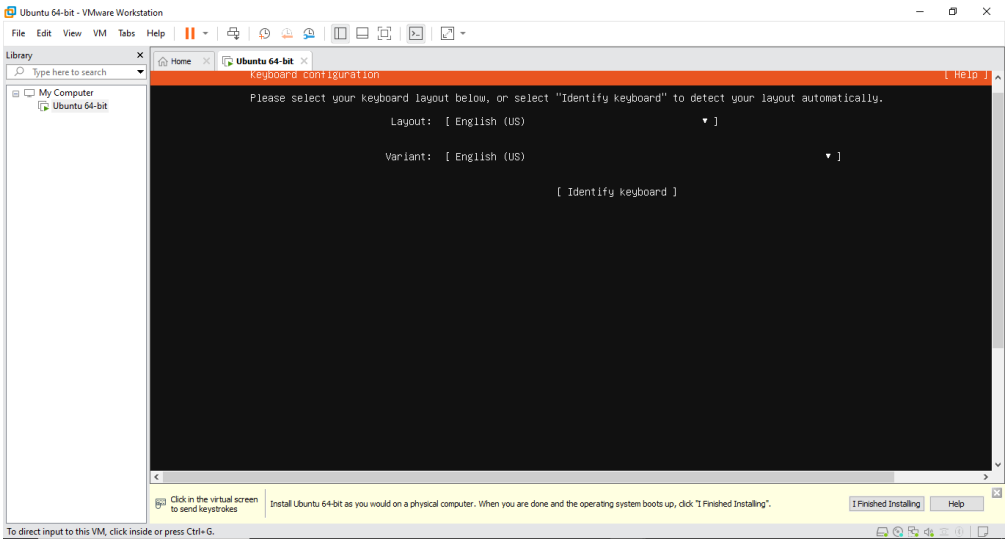


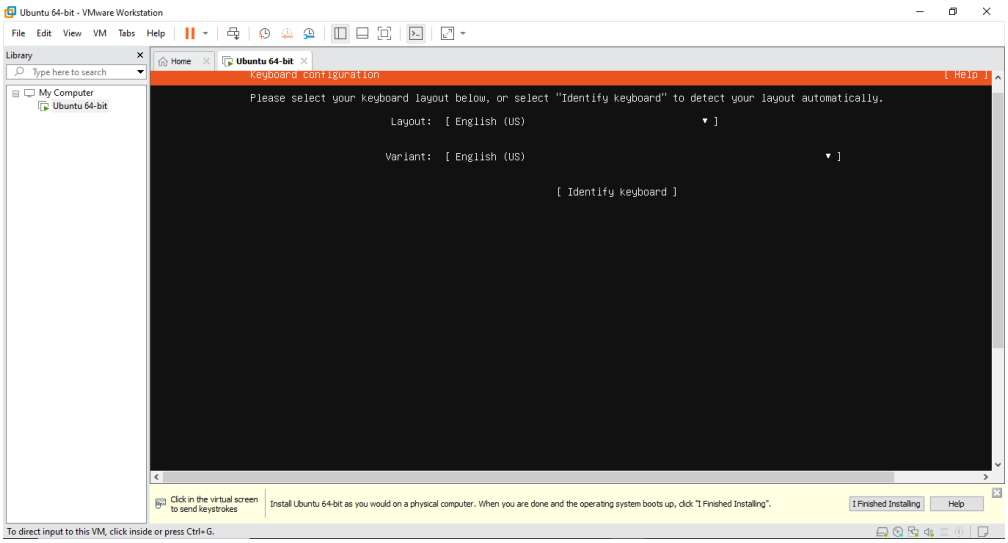


Task 5: Choose the Installation Language

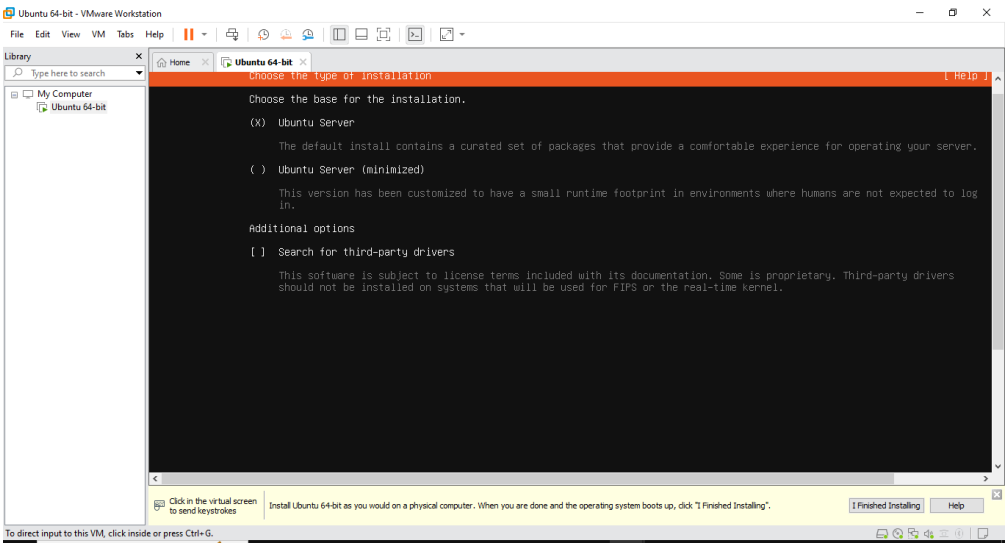


Task 6: Configure the Keyboard Layout

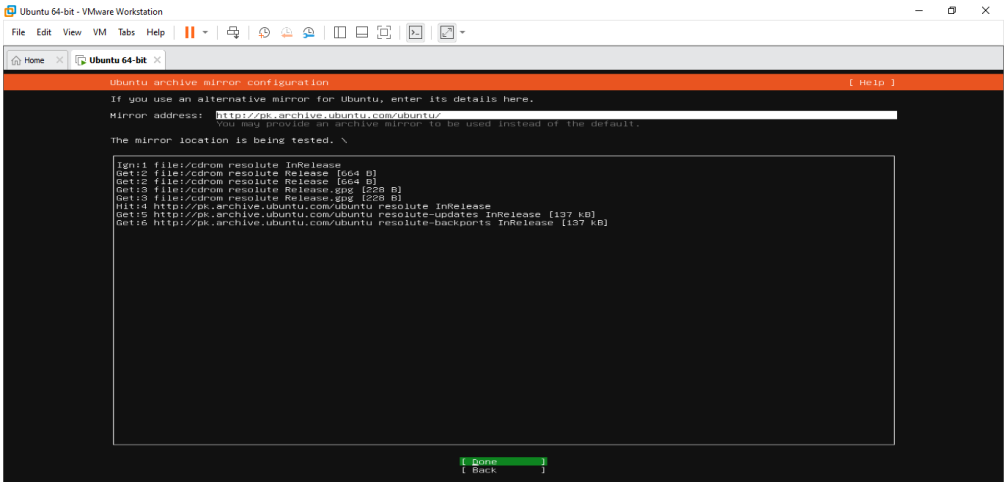


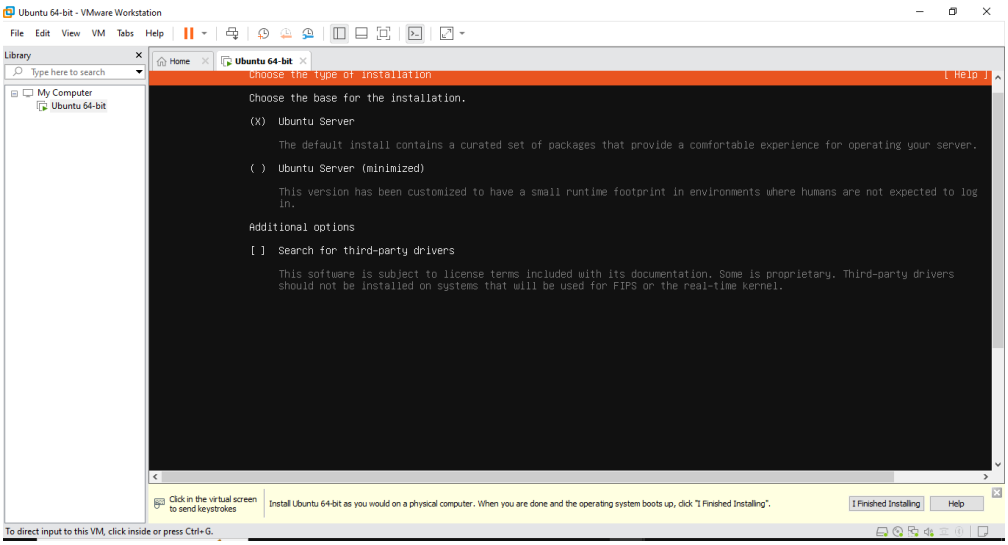


Task 7: Choose the Ubuntu Server Installation Type

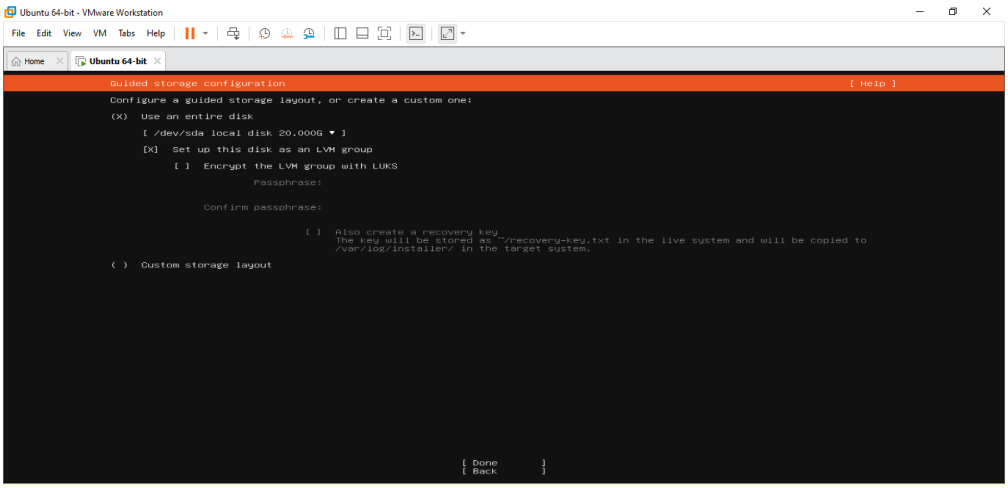


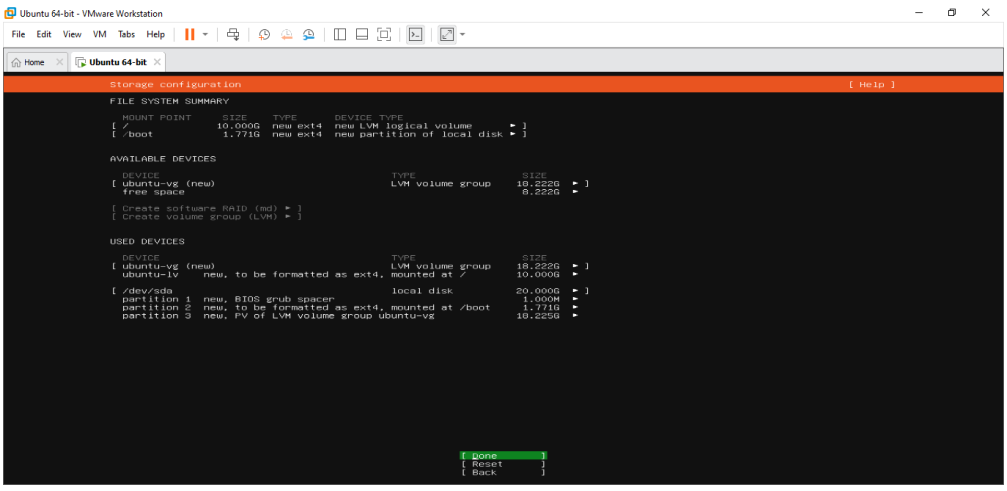
Task 8: Configure the Network

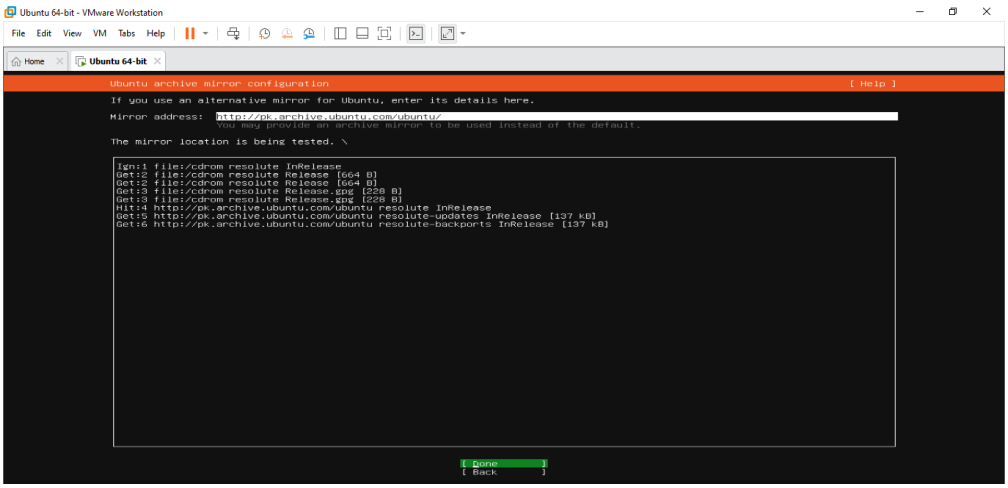




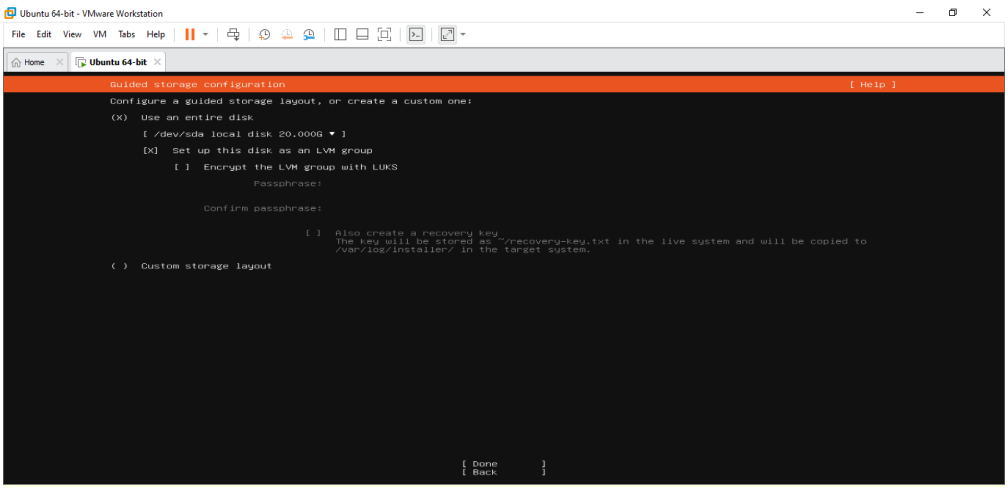
Task 9: Configure Storage


Ubuntu 64-bit - VMware Workstation
File Edit View VM Tabs Help
Home
Ubuntu 64-bit
Guided storage configuration [Help]
Configure a guided storage layout, or create a custom one:
(X) Use an entire disk
[/dev/sda local disk 20.000G ▾]
[X] Set up this disk as an LVM group
[ ] Encrypt the LVM group with LUKS
Passphrase:
Confirm passphrase:
[ ] Also create a recovery key
The key will be stored as ~/recovery-key.txt in the live system and will be copied to
/var/log/installer/ in the target system.
( ) Custom storage layout
[Done]
[Back]

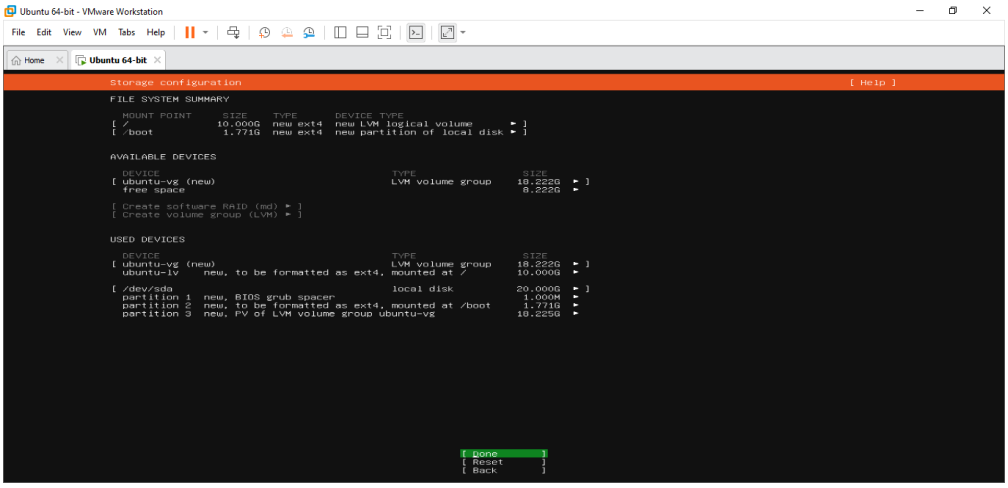

Ubuntu 64-bit - VMware Workstation
File Edit View VM Tabs Help
Home
Ubuntu 64-bit
Storage configuration [Help]
FILE SYSTEM SUMMARY
MOUNT POINT SIZE TYPE DEVICE TYPE
[/ 10.000G new ext4 new LVM logical volume ▸]
[/boot 1.771G new ext4 new partition of local disk ▸]
AVAILABLE DEVICES
DEVICE TYPE SIZE
[ubuntu-vg (new) LVM volume group 18.222G ▸]
free space 8.222G ▸
[Create software RAID (md) ▸]
[Create volume group (LVM) ▸]
USED DEVICES
DEVICE TYPE SIZE
[ubuntu-vg (new) LVM volume group 18.222G ▸]
ubuntu-lv new, to be formatted as ext4, mounted at / 10.000G ▸
[/dev/sda local disk 20.000G ▸]
partition 1 new, BIOS grub spacer 1.000M ▸
partition 2 new, to be formatted as ext4, mounted at /boot 1.771G ▸
partition 3 new, PV of LVM volume group ubuntu-vg 10.225G ▸
[Done]
[Reset]
[Back]


Ubuntu 64-bit - VMware Workstation
File Edit View VM Tabs Help
Home
Ubuntu 64-bit
Ubuntu archive mirror configuration [Help]
If you use an alternative mirror for Ubuntu, enter its details here.
Mirror address: http://pk.archive.ubuntu.com/ubuntu/
You may provide an archive mirror to be used instead of the default.
The mirror location is being tested. \
Ign:1 file:/cdrom resolute InRelease
Get:2 file:/cdrom resolute Release [664 B]
Get:2 file:/cdrom resolute Release [664 B]
Get:3 file:/cdrom resolute Release.gpg [228 B]
Get:3 file:/cdrom resolute Release.gpg [228 B]
Hit:4 http://pk.archive.ubuntu.com/ubuntu resolute InRelease
Get:5 http://pk.archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:6 http://pk.archive.ubuntu.com/ubuntu resolute-backports InRelease [137 kB]
[Done]
[Back]

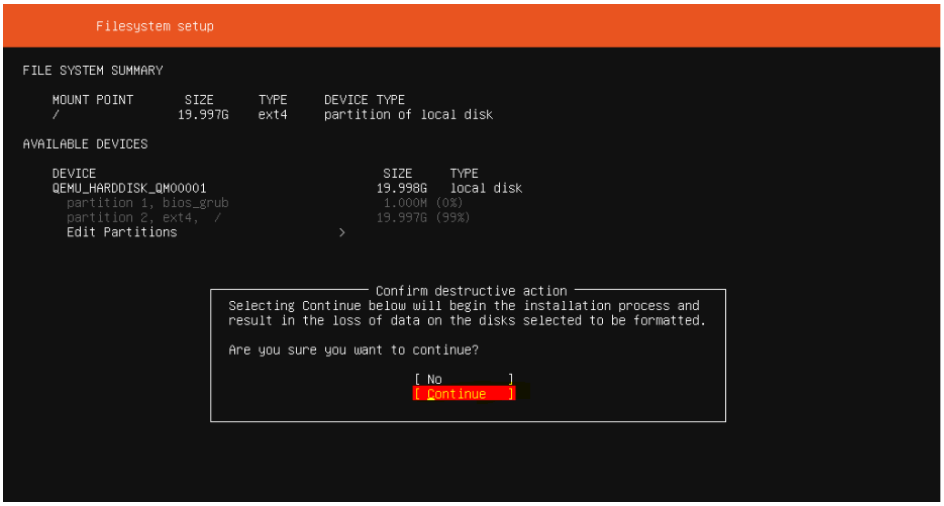
Task 10: Select the Installation Disk



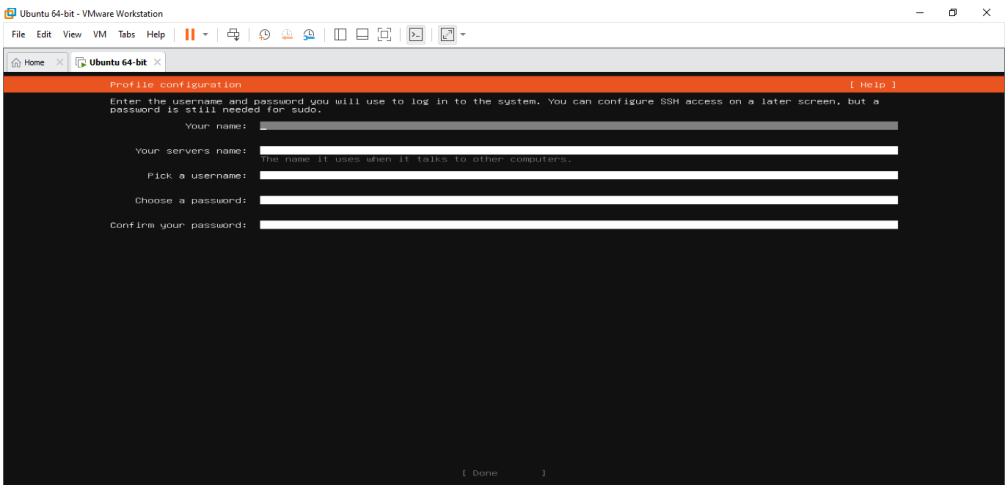
Task 11: Review the Partitions

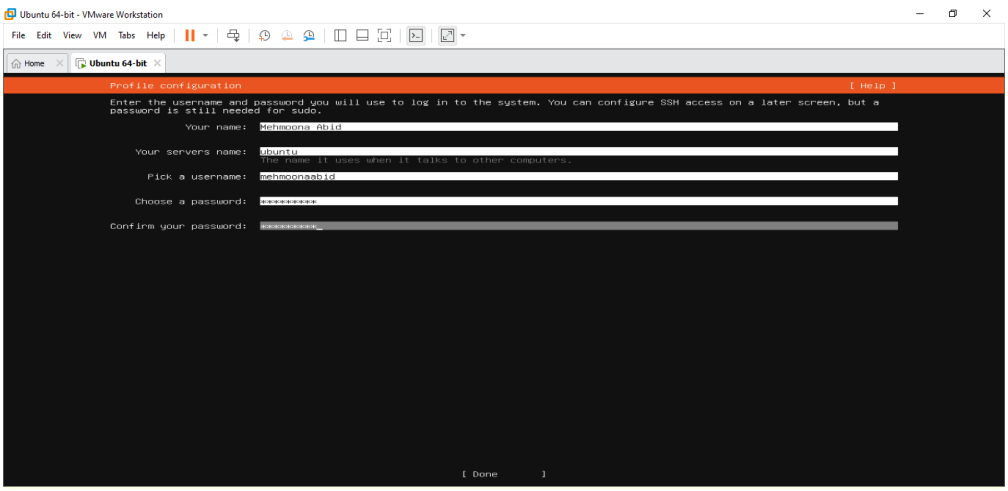


Task 12: Confirm the Storage Changes

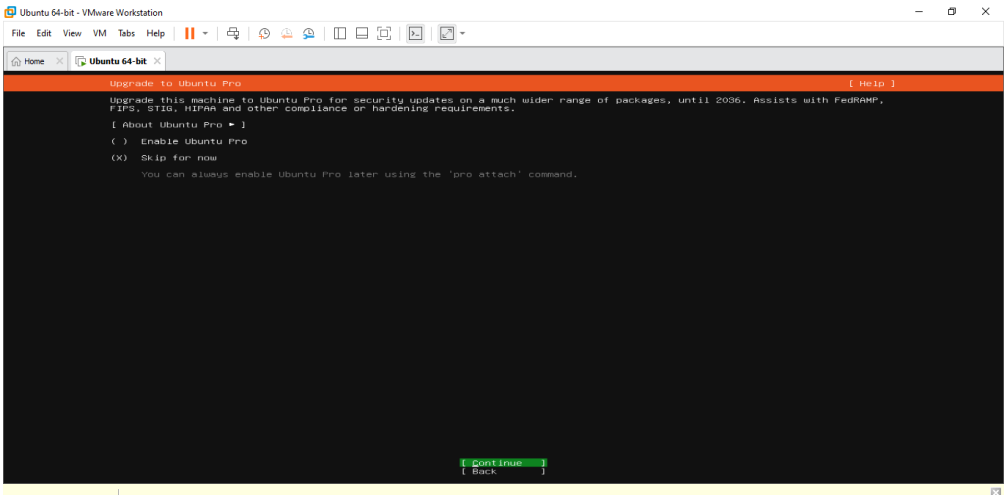


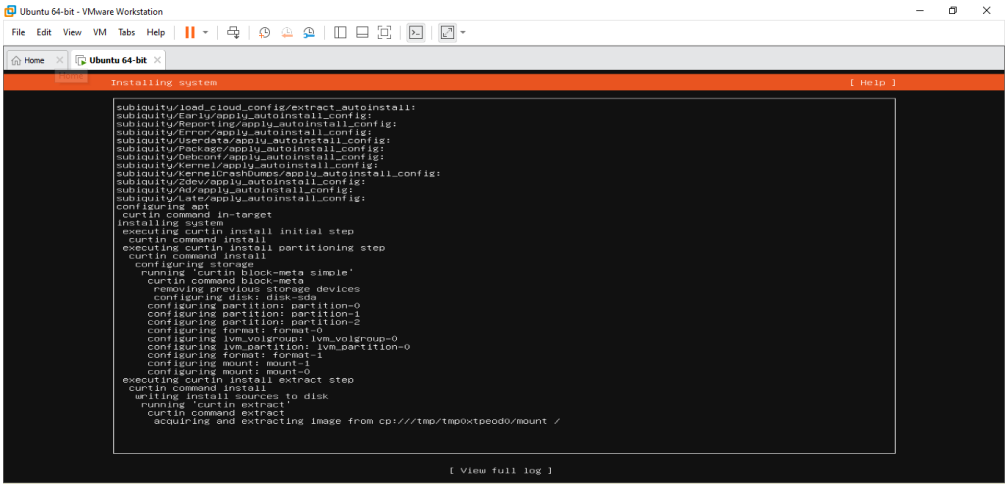
Task 13: Configure the Ubuntu Profile



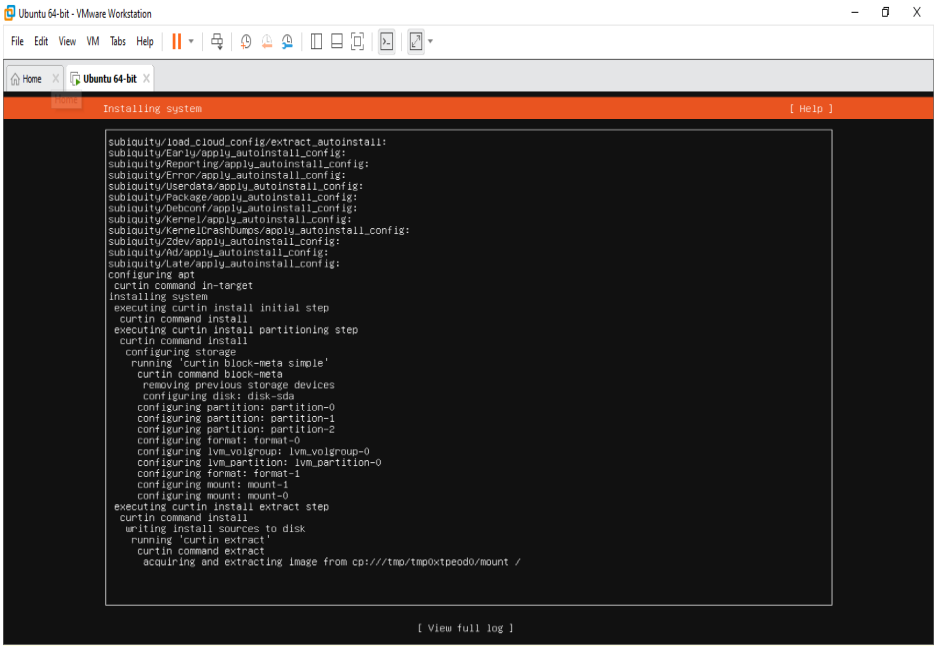


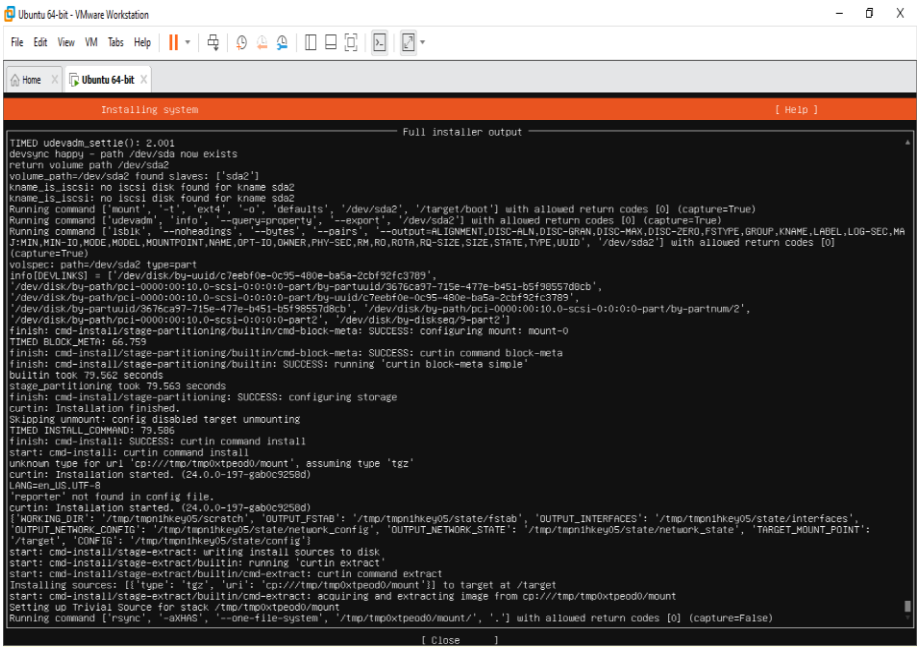
Task 14: Complete the Software Installation

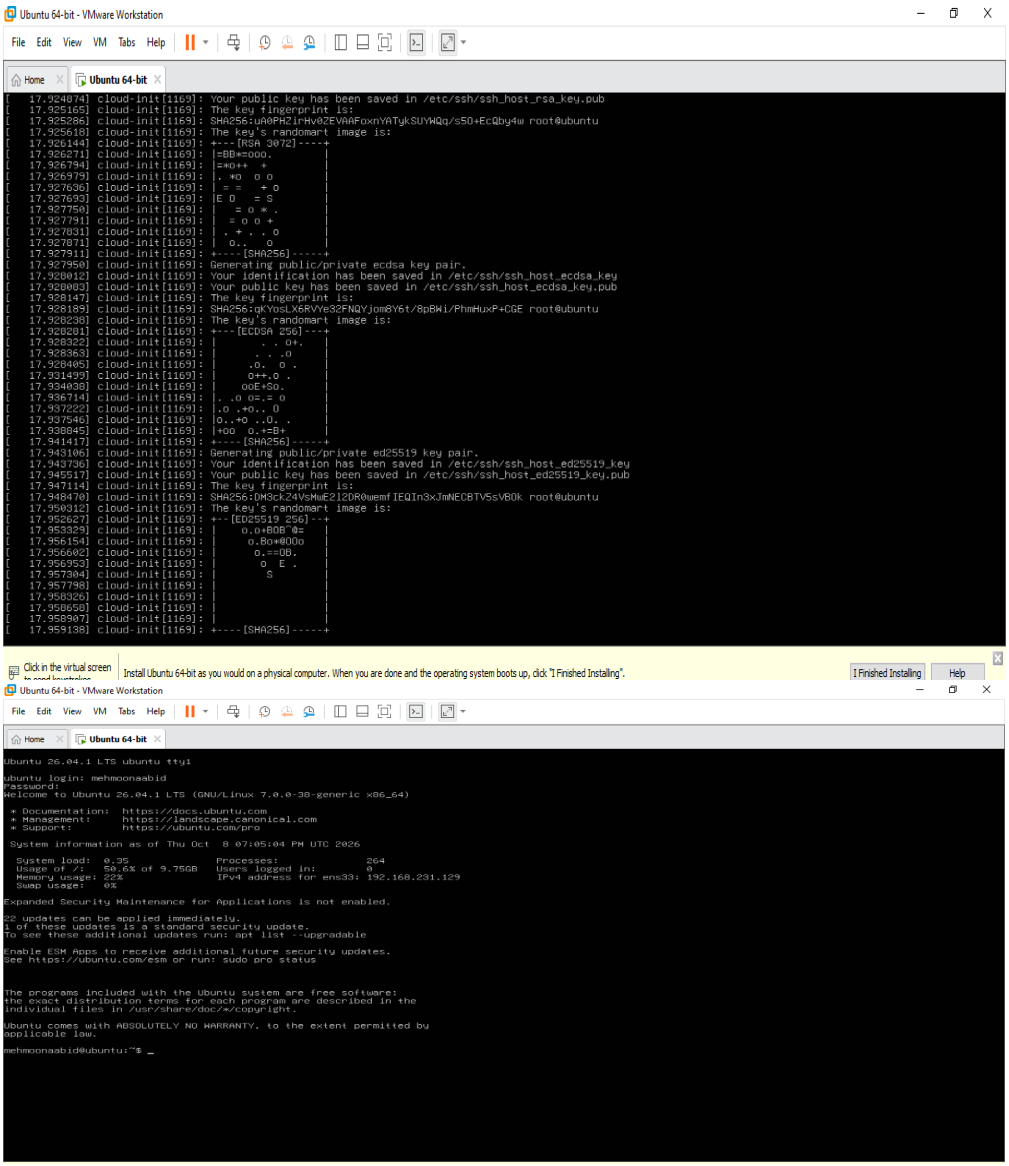

Ubuntu 64-bit - VMware Workstation
File Edit View VM Tabs Help
Home
Ubuntu 64-bit
Upgrade to Ubuntu Pro
[Help]
Upgrade this machine to Ubuntu Pro for security updates on a much wider range of packages, until 2036. Assists with FedRAMP, FIPS, STIG, HIPAA and other compliance or hardening requirements.
[About Ubuntu Pro ►]
( ) Enable Ubuntu Pro
(X) Skip for now
You can always enable Ubuntu Pro later using the 'pro attach' command.
[Continue]
[Back]


Ubuntu 64-bit - VMware Workstation
File Edit View VM Tabs Help
Home
Ubuntu 64-bit
Installing system
[Help]
subiquity/load_cloud_config/extract_autoinstall:
subiquity/Early/apply_autoinstall_config:
subiquity/Reporting/apply_autoinstall_config:
subiquity/Error/apply_autoinstall_config:
subiquity/Userdata/apply_autoinstall_config:
subiquity/Package/apply_autoinstall_config:
subiquity/Debconf/apply_autoinstall_config:
subiquity/Kernel/apply_autoinstall_config:
subiquity/KernelCrashDumps/apply_autoinstall_config:
subiquity/Zdev/apply_autoinstall_config:
subiquity/Ad/apply_autoinstall_config:
subiquity/Late/apply_autoinstall_config:
configuring apt
curtin command in-target
installing system
executing curtin install initial step
curtin command install
executing curtin install partitioning step
curtin command install
configuring storage
running 'curtin block-meta simple'
curtin command block-meta
removing previous storage devices
configuring disk: disk-sda
configuring partition: partition-0
configuring partition: partition-1
configuring partition: partition-2
configuring format: format-0
configuring lvm_volgroup: lvm_volgroup-0
configuring lvm_partition: lvm_partition-0
configuring format: format-1
configuring mount: mount-1
configuring mount: mount-0
executing curtin install extract step
curtin command install
writing install sources to disk
running 'curtin extract'
curtin command extract
acquiring and extracting image from cp:///tmp/tmp0xtpeod0/mount /
[View full log]

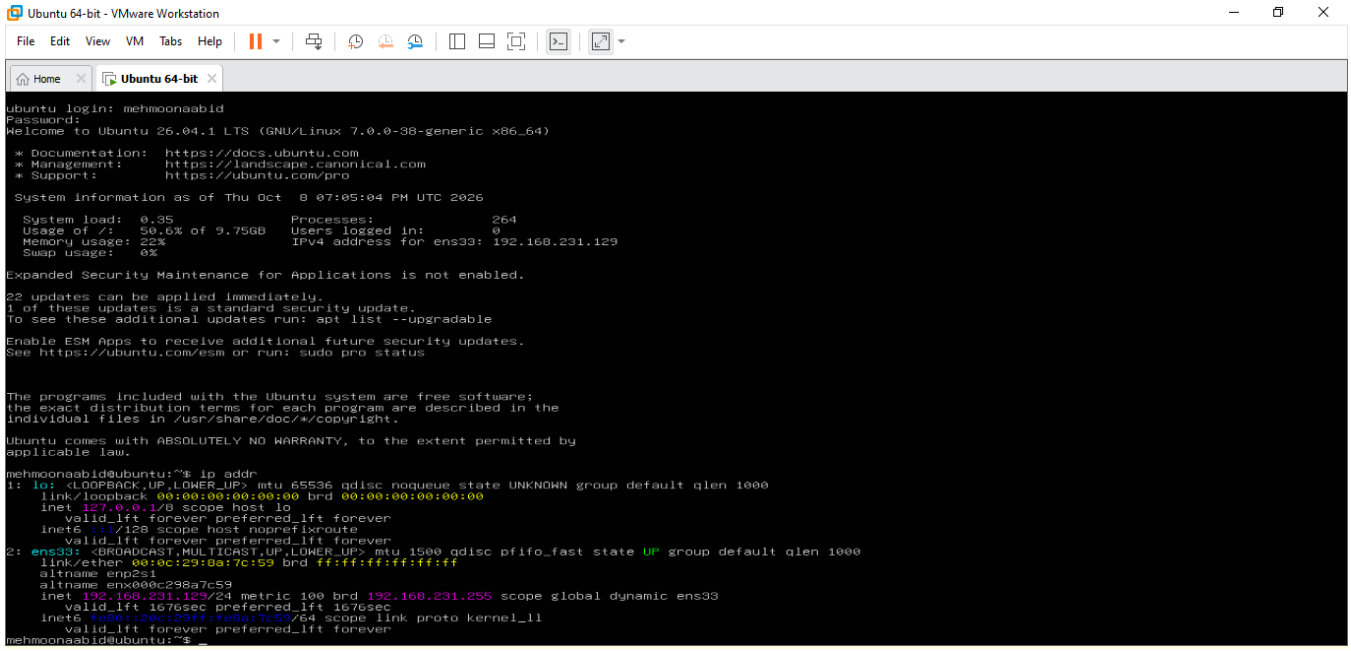
Task 15: Reboot and Verify the Ubuntu Identity

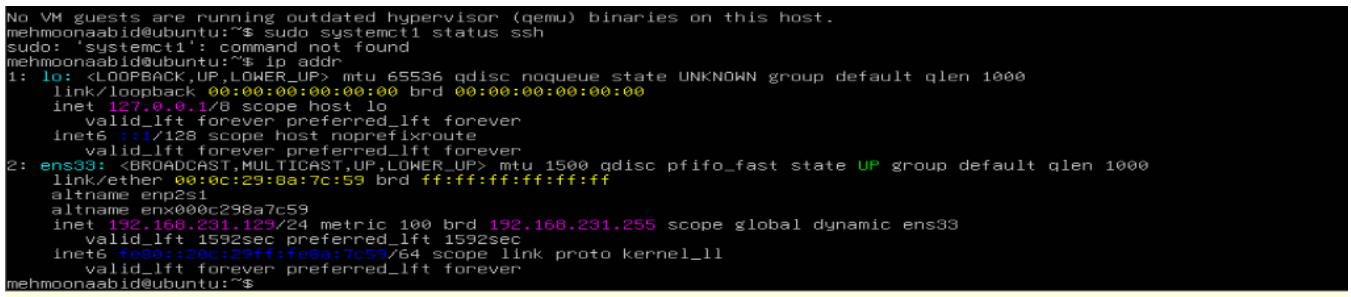


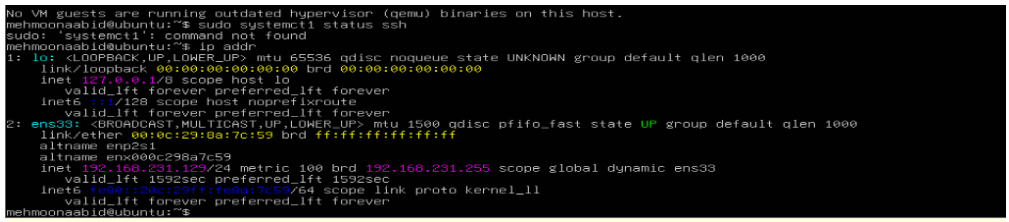




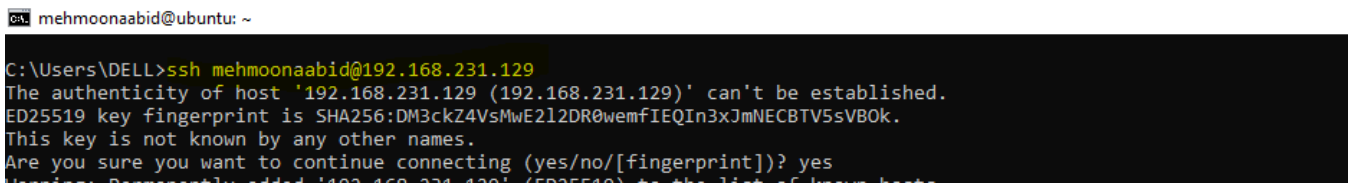
Task 16: Find the Ubuntu Server IP Address







Task 17: Connect to Ubuntu Server from Windows



mehmoonaabid@ubuntu: ~

```
C:\Users\DELL>ssh mehmoonaabid@192.168.231.129
The authenticity of host '192.168.231.129 (192.168.231.129)' can't be established.
ED25519 key fingerprint is SHA256:DM3ckZ4VsMwE2l2DR0wemfIEQIn3xJmNECBTV5sVBOk.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
```

```
C:\Users\DELL>ssh mehmoonaabid@192.168.231.129
The authenticity of host '192.168.231.129 (192.168.231.129)' can't be established.
ED25519 key fingerprint is SHA256:DM3ckZ4VsMwE2l2DR0wemfIEQIn3xJmNECBTV5sVBOk.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.231.129' (ED25519) to the list of known hosts.
mehmoonaabid@192.168.231.129's password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-38-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Thu Oct  8 07:26:11 PM UTC 2026

  System load:  0.04              Processes:             221
  Usage of /:   50.8% of 9.75GB   Users logged in:       0
  Memory usage: 25%               IPv4 address for ens33: 192.168.231.129
  Swap usage:   0%


Expanded Security Maintenance for Applications is not enabled.

22 updates can be applied immediately.
1 of these updates is a standard security update.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


mehmoonaabid@ubuntu:~$
```